Real Estate Valuation

STATISTICAL LEARNING

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Abstract

This report analyzes residential property values in Taipei, Taiwan, using the *Real Estate Valuation* dataset from the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/477/real+estate+valuation+data+set).

The study first explores the structure of the housing market through Principal Component Analysis (PCA) and clustering, revealing three economically meaningful groups of properties largely differentiated by accessibility to MRT stations, neighborhood amenities, and building age.

A supervised learning framework is then applied to assess the predictive power of structural and locational features on house prices. Among the models considered—linear regression, Neural Networks, and Random Forest—the latter provides the best performance, confirming the importance of non-linear spatial effects.

Overall, the results highlight that proximity to MRT stations and geographical location are the strongest determinants of housing prices, while structural attributes such as house age play a secondary role. These findings illustrate both the segmentation of Taipei's housing market and the relevance of flexible models for price prediction.

1. Dataset Description and Preprocessing

The analysis is based on the *Real Estate Valuation Data Set* (Yeh, 2018), publicly available from the UCI Machine Learning Repository. The dataset contains 414 observations of residential housing transactions in Taipei, Taiwan, collected between 2012 and 2013. Each record includes the transaction price per unit area (measured in New Taiwan Dollars per square meter) and several structural and locational characteristics.

The variables used in this project are:

- **TransactionDate**: the year and month of the transaction, expressed as a decimal (e.g., 2013.250 ≈ March 2013).
- **HouseAge**: the age of the property in years.
- **DistanceMRT**: the distance to the nearest mass rapid transit (MRT - metro) station, measured in meters.
- **ConvenienceStores**: the number of convenience stores in the living circle on foot.
- **Latitude and Longitude**: geographical coordinates of the property location.
- **PricePerUnitArea**: the target variable, defined as the unit price of the property (10,000 New Taiwan Dollar/Ping, where Ping is a local unit, 1 Ping = 3.3 meter squared).

Preprocessing

The dataset was subjected to preliminary cleaning before the analysis:

1. **Missing values**: no missing values were found across the variables.
2. **Duplicated entries**: no duplicates were detected.
3. **Variable types**: all variables are numerical and therefore directly suitable for multivariate statistical analysis.
4. **TransactionDate:** this variable was not incorporated into the predictive models and was thus not subjected to additional transformation.
5. **Unit conversion:** the variable *PricePerUnitArea* was converted from 10,000 New Taiwan Dollar per Ping into 10,000 New Taiwan Dollar per m² to allow for standard interpretation and comparison.

Overall, the dataset is clean, well-structured, and ready for application of both unsupervised and supervised learning techniques.

2. Key findings

- Three market segments emerge from PCA and clustering, mainly differentiated by proximity to MRT stations, housing age, and the availability of local amenities.
- Geographical location and distance to MRT stations are the strongest predictors of house prices, consistently dominating feature importance across models.
- Random Forest achieves the best predictive performance, outperforming linear model and the neural network, suggesting the relevance of non-linear spatial relationships.

3. Methodology

The methodological approach of this project combines unsupervised and supervised statistical learning techniques. The goal is to explore the structure of the housing market in Taipei and evaluate the predictive power of different models for housing prices.

3.1 Unsupervised Learning

Unsupervised learning methods are used to identify hidden structures in the dataset without relying on predefined labels. Two techniques are applied:

- **Principal Component Analysis (PCA):** used to reduce dimensionality and summarize the variance of the data into a smaller set of uncorrelated components. This step helps to highlight the most influential variables driving heterogeneity among houses.
- **Clustering (k-means and hierarchical):** applied on the transformed feature space to detect homogeneous groups of properties. Cluster solutions are evaluated using criteria such as the elbow method and silhouette scores.

3.2 Supervised Learning

Supervised learning methods are employed to predict the target variable (*PricePerUnitArea*) based on structural and locational features. Several models are compared:

- **Linear Regression:** baseline model, providing a simple interpretable benchmark.
- **Artificial Neural Network (ANN):** a simple feed-forward neural network (multilayer perceptron), included to test the performance of non-linear models inspired by machine learning.

- **Random Forest:** a tree-based ensemble method that captures nonlinear relationships and interactions among predictors.

3.3 Model Evaluation

For the supervised learning task, the dataset is split into training and test sets. Models are evaluated using out-of-sample performance metrics:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Coefficient of Determination (R^2)

These measures allow for a fair comparison of predictive accuracy across traditional statistical models and modern machine learning approaches.

4. Exploratory Data Analysis (EDA)

Before applying unsupervised and supervised learning methods, an exploratory data analysis was performed to examine the distribution of the target variable and its relationships with the main predictors.

4.1 Summary Statistics

Table 4.1 reports descriptive statistics for the variables used in the analysis.

The average unit price of housing is 115,000 NTD/m², with a median of 117,000 NTD/m², although values range widely (from 23,000 to 356,000 NTD/m²), indicating substantial price heterogeneity across properties.

Housing age displays a mean of 17.7 years, with a relatively wide range (0–43.8), suggesting the coexistence of both recently built and older structures.

Distance to the nearest MRT station also shows strong dispersion, with an average of 1,084 meters and a median of 492 meters, reflecting a right-skewed distribution and highlighting that many properties enjoy good accessibility, while a non-negligible group is located farther away (up to ~6.5 km).

The number of convenience stores within walking distance varies between 0 and 10, with an average of 4.1, pointing to meaningful differences in local amenities.

Table 4.1 — Summary Statistics

Variable	Mean	Median	SD	Min	Max
PricePerUnitArea (10,000 NTD/m²)	11.5	11.7	4.12	2.30	35.6
HouseAge	17.7	16.1	11.4	0.0	43.8
DistanceMRT	1084	492	1262	23.4	6488
ConvenienceStores	4.09	4	2.95	0	10

4.2 Distribution of House Prices

The distribution of housing prices per square meter appears highly right-skewed, with most properties concentrated between 50,000 and 150,000 NTD per m², and a few extreme high-price

outliers (Figure 4.1). This asymmetry suggests that a log-transformation may be required for subsequent modeling.

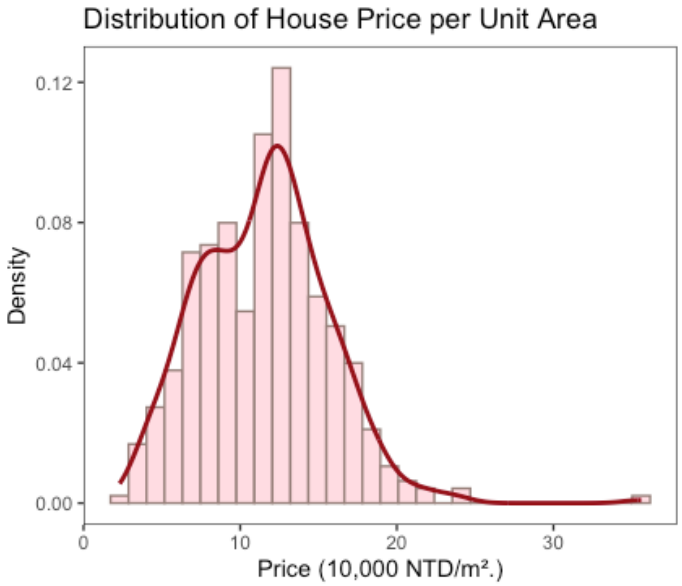

Figure 4.1. Distribution of House Price per Unit Area

After applying the logarithmic transformation, the price distribution becomes more symmetric (Figure 4.2). This transformation reduces the impact of extreme outliers and stabilizes variance, making the data more suitable for regression and other supervised learning models.

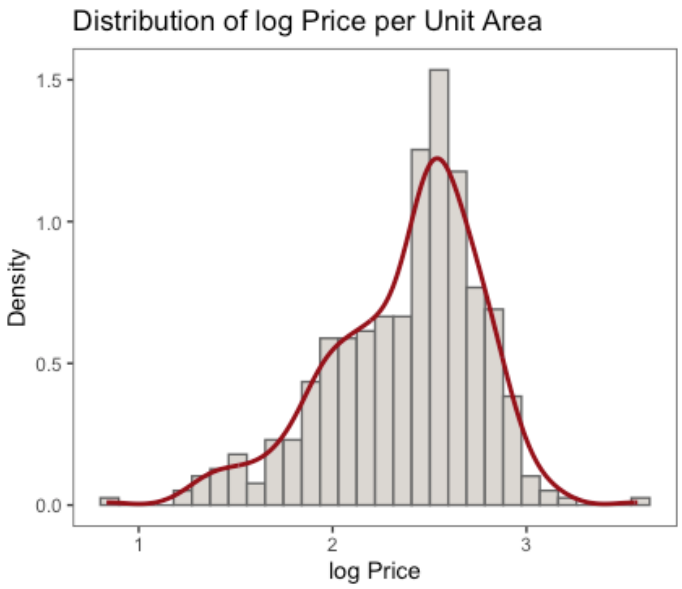

Figure 4.2. Distribution of log(Price per Unit Area)

4.3 Relationship with Location and Amenities

There is a clear negative relationship between housing prices and distance to the nearest MRT station (Figure 4.3). Properties located closer to public transport hubs tend to have higher unit prices, highlighting the central role of accessibility in urban real estate valuation.

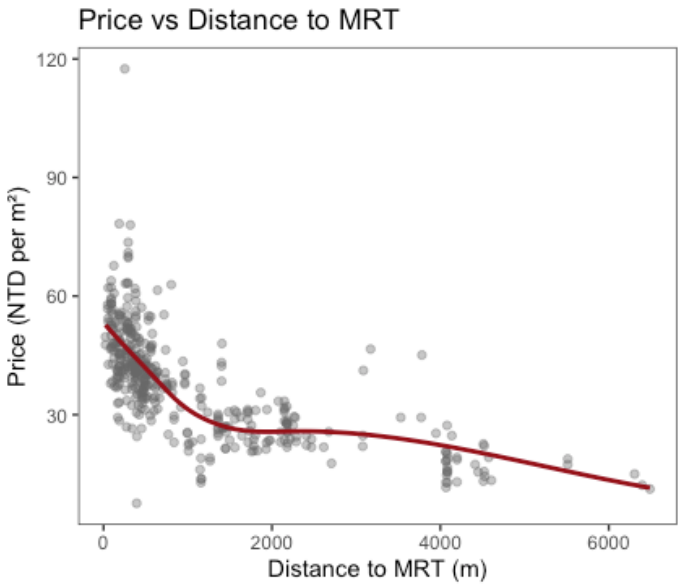

Figure 4.3. Price vs Distance to MRT

A positive association emerges between housing prices and the number of nearby convenience stores (Figure 4.4). Properties located in areas with a higher density of amenities are systematically more expensive, confirming the importance of neighborhood characteristics.

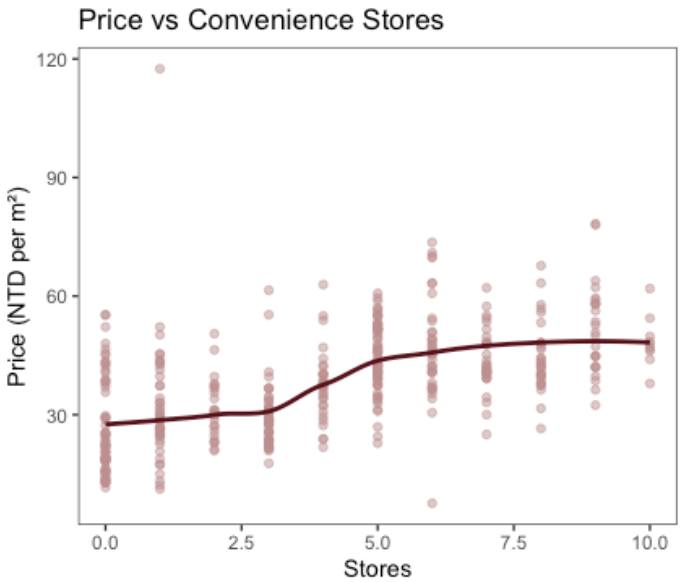

Figure 4.4. Price vs Number of Convenience Stores

4.4 Correlation Structure

The correlation matrix (Figure 4.5) confirms the main bivariate patterns: unit price is negatively correlated with distance to MRT (−0.67) and positively correlated with both convenience stores (+0.57) and geographical coordinates (latitude and longitude). These results support the idea that location and neighborhood services are key determinants of housing prices in the dataset.

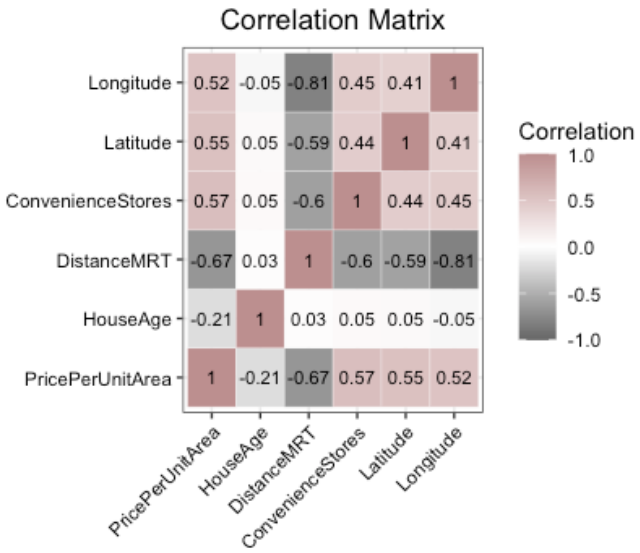

Figure 4.5. Correlation Matrix of Numerical Variables

5. PCA Results

The Principal Component Analysis (PCA) was applied to five predictors (*HouseAge*, *DistanceMRT*, *ConvenienceStores*, *Latitude*, and *Longitude*). The eigenvalue decomposition (Figure 5.1) shows that the first two principal components explain 74% of the total variance (PC1: 53.5%, PC2: 20.4%). Given this high cumulative proportion, the analysis focuses on PC1 and PC2.

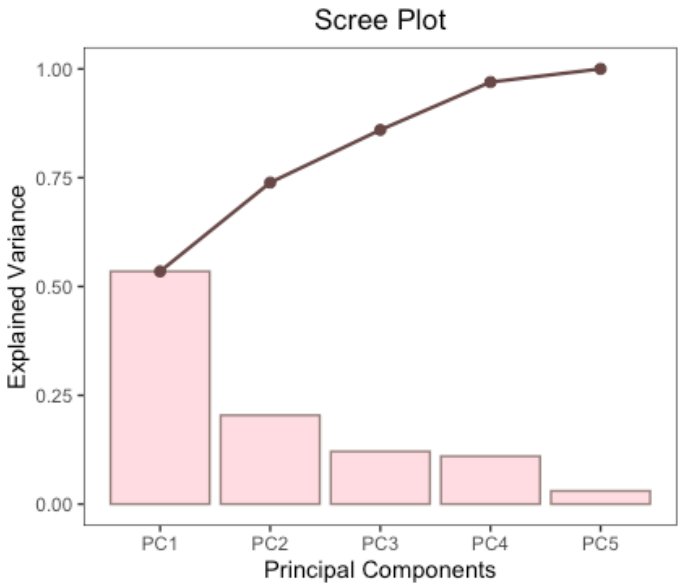

Figure 5.1. Scree Plot

The loadings highlight the role of each variable in shaping the components (Table 1).

PC1 contrasts *DistanceMRT* (positive loading) with *ConvenienceStores*, *Latitude*, and *Longitude* (negative loadings). This axis can therefore be interpreted as a dimension of accessibility and geographical location.

PC2 is dominated by *HouseAge* (loading = 0.975), representing an age-related dimension almost entirely independent from locational features.

Table 5.1. Loadings on the first two principal components

Variable	PC1	PC2
HouseAge	-0.005	0.975
DistanceMRT	0.570	0.065
ConvenienceStores	-0.461	0.102
Latitude	-0.450	0.125
Longitude	-0.509	-0.138

The correlation circle (Figure 5.2) illustrates these relationships: *HouseAge* aligns closely with PC2, while *DistanceMRT* extends in the positive PC1 direction, opposed to *Latitude, Longitude,* and *ConvenienceStores*. The distance of the arrows from the origin reflects the strength of representation on the PC1–PC2 plane. Angles between vectors reveal correlations among variables: for example, *Latitude* and *Longitude* are strongly associated, while both contrast with *DistanceMRT*.

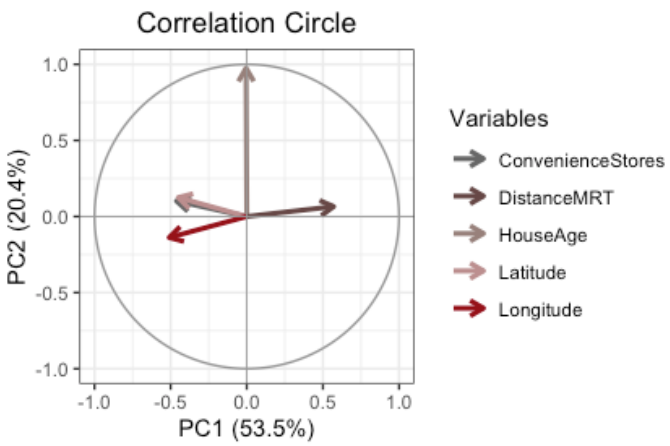

Figure 5.2. PCA Correlation Circle

The score plot (Figure 5.3) shows the distribution of housing transactions in the new PC space. Along PC1, properties separate according to transport accessibility and availability of convenience stores, whereas along PC2, properties differ primarily in terms of age. The presence of discernible clusters in the scatter suggests that the dataset may naturally divide into groups of houses with similar structural and locational characteristics, providing a suitable basis for subsequent clustering analysis.

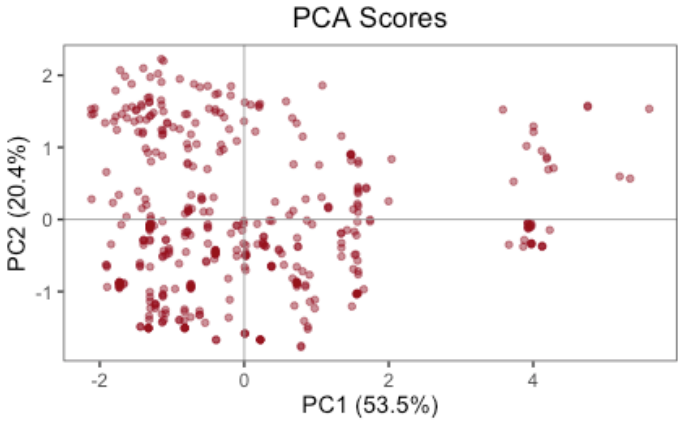

Figure 5.3. PCA Scores

6. Clustering Results

To explore potential segmentation in the housing market, k-means clustering was applied on the first three principal components of the dataset. While the interpretation of PCA focused primarily on PC1 and PC2, clustering on the first three PCs (explaining approximately 86% of total variance) was adopted to increase the robustness of the grouping structure.

Two complementary approaches were used to determine the appropriate number of clusters: the Elbow Method and the Silhouette Analysis.

- The Elbow plot of the within-cluster sum of squares (WSS) (Figure 6.1) suggested a possible inflection around $k = 4 - 5$

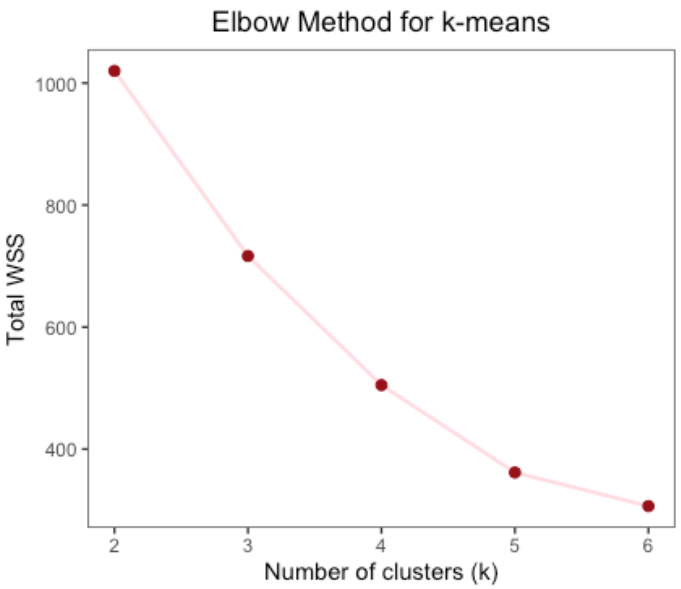

Figure 6.1. Elbow Method for k-means clustering

- The average silhouette width (Figure 6.2) reached its maximum at $k = 5$, indicating a statistically stronger separation between clusters.

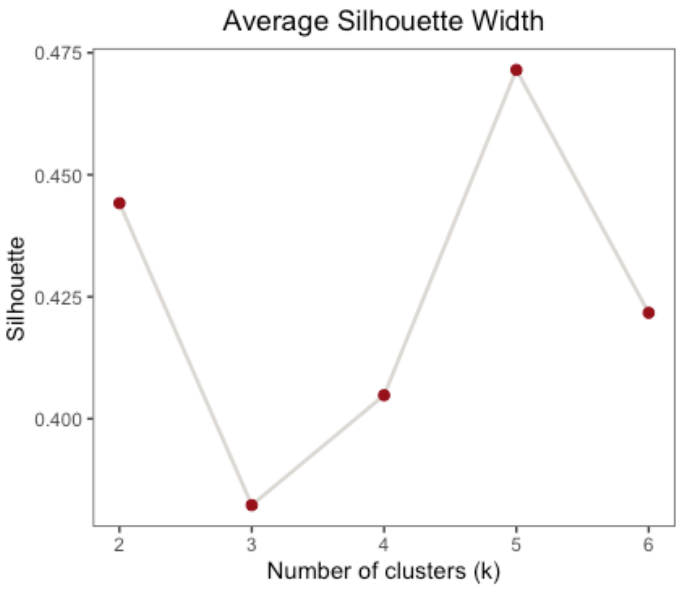

Figure 6.2. Average silhouette width across different k values

However, in line with the principle of parsimony and interpretability, $k = 3$ was retained as the final solution. This choice balances explanatory power with clarity, producing three economically meaningful groups of properties.

Table 6.1. Cluster Profiles

Cluster	n	Mean Price	Mean Age	Mean Distance MRT	Mean Stores
1	102	12.2	33.3	460	5.85
2	90	6.92	18.1	2991	1.52
3	222	13.0	10.4	597	4.33

- **Cluster 1**: older houses (≈ 33 years), close to MRT stations (≈ 460 m), with a high density of convenience stores (≈ 6). Prices are moderately high (≈ 122,000 NTD/m²). Probably correspond to central areas.
- **Cluster 2**: properties located far from MRT stations (≈ 3 km) and with few nearby stores (≈ 1.5). Houses in this group have a medium age (≈ 18 years) and probably correspond to peripheral areas, with the lowest average prices (≈ 69,200 NTD/m²).
- **Cluster 3**: the youngest houses (≈ 10 years), relatively well connected (≈ 600 m from MRT), with an intermediate level of amenities (≈ 4 stores). This cluster records the highest average prices (≈ 130,000 NTD/m²).

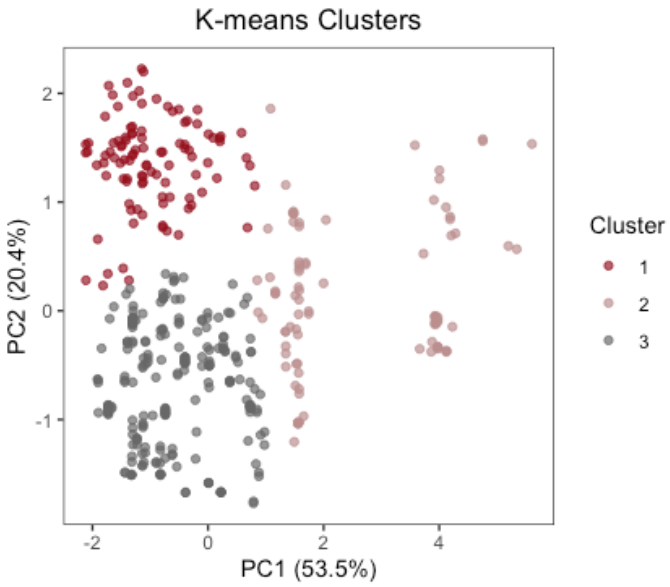

Figure 6.3. K-means clustering results (k = 3) projected onto PC1–PC2. Clustering was performed on the first three PCs.

Overall, the clustering analysis highlights three distinct market segments: peripheral low-price houses, central older houses with moderate prices, and central newer houses commanding premium prices. These results confirm the joint role of accessibility, neighborhood amenities, and building age in shaping price heterogeneity in Taipei's housing market.

These results provide a clear segmentation of the Taipei housing market, which serves as a foundation for the next step of the analysis: evaluating the predictive power of supervised learning models for housing prices.

7. Data Preparation for Supervised Learning

Before fitting predictive models, the dataset was prepared for regression analysis. The target variable was defined as the natural logarithm of the house price per unit area (*LogPrice*), in order to reduce skewness and stabilize variance. The predictors included structural and locational variables: house age, distance to the nearest MRT station, number of convenience stores, latitude, and longitude.

The dataset was randomly split into a training set (70%) and a test set (30%), ensuring reproducibility by fixing the random seed. To prevent data leakage, standardization of predictors (mean zero, unit variance) was performed using only the parameters from the training set and then applied to the test set. This scaling was applied for models sensitive to the magnitude of predictors (linear regression, and neural networks), while raw predictors were retained for tree-based methods such as Random Forest.

This procedure ensured comparability across models and provided a robust basis for evaluating predictive performance.

8. Baseline Model: Linear Regression

As a starting point, a linear regression model was estimated on the training data using the standardized predictors. Linear regression provides an interpretable benchmark against which more flexible models can be compared.

Model performance was assessed using R^2 and Root Mean Squared Error (RMSE) on both the training and test sets. Results are reported in Table 8.1.

Table 8.1. Performance of the Baseline Linear Regression Model

Dataset	R^2	RMSE
Train	0.713	0.198
Test	0.604	0.275

The model explains approximately 71% of the variance in the training set and 60% in the test set, with only a moderate increase in prediction error out-of-sample, suggesting acceptable generalization.

Estimated coefficients (in Table 8.2) confirm expected relationships between structural and locational predictors:

- House age (−0.063) and distance to MRT (−0.162) show negative effects on log prices. Interpreting coefficients on the log scale, this means that each additional year of age reduces the expected price by about 6%, and increasing distance from MRT stations lowers housing values.
- Convenience stores (+0.076) have a positive impact: each additional nearby store is associated with an increase of about 7.6% in housing prices, confirming the role of neighborhood amenities.
- Geographical coordinates (latitude +0.108, longitude +0.023) are positively associated with log prices. These coefficients do not represent causal effects but rather capture the spatial dimension of the Taipei housing market.

From a geographical perspective, the most expensive districts of Taipei (such as Xinyi, the financial center in the east, and Da'an, a central cultural district) are located in the eastern and north-eastern parts of the city, which roughly correspond to higher values of latitude and longitude in the dataset. This suggests that the positive coefficients on coordinates reflect the concentration of high-price areas in those locations.

Table 8.2. Model Coefficients

Variable	Coefficient
(Intercept)	2.353
HouseAge	-0.063
DistanceMRT	-0.162
ConvenienceStores	0.076
Latitude	0.108
Longitude	0.023

9. Neural Network (MLP)

To explore potential nonlinear relationships in the data, a Multilayer Perceptron (MLP) neural network was trained on the standardized predictors. The architecture consists of a single hidden layer with 5 neurons, optimized via backpropagation with a weight decay of 0.01 to control overfitting. The network was trained for up to 500 iterations.

Model performance, summarized in Table 9.1, shows that the MLP achieves an R^2 of 0.85 on the training set and 0.66 on the test set. Compared to the linear and Ridge regressions, the neural network yields slightly higher predictive accuracy on unseen data, indicating that some degree of nonlinearity in the relationships between structural and locational variables is being captured.

Table 9.1. Performance of Neural Network (MLP)

Dataset	R^2	RMSE
Train	0.845	0.145
Test	0.659	0.256

Despite this improvement, the gain in performance remains moderate.

10. Random Forest

A Random Forest (RF) regressor was trained on the raw predictors, as tree-based models do not require feature standardization.

The baseline model used 500 trees and sampled two variables (`mtry = 2`) at each split, with variable-importance computation enabled.

Table 10.1. Random Forest Performance

Dataset	R^2	RMSE
Train	0.951	0.081
Test	0.703	0.238

The Random Forest captures non-linear relationships and interactions among predictors, yielding higher test accuracy than the linear models.

10.2 Variable Importance

Figure 10.1 reports the variable importance from the Random Forest model based on IncNodePurity. Distance to the nearest MRT station is by far the most influential predictor, confirming that proximity to public transportation is a key driver of housing prices in Taipei. Geographic coordinates (latitude and longitude) follow, highlighting the relevance of spatial location. Structural features such as house age and number of convenience stores contribute less to predictive accuracy, though they still provide useful information.

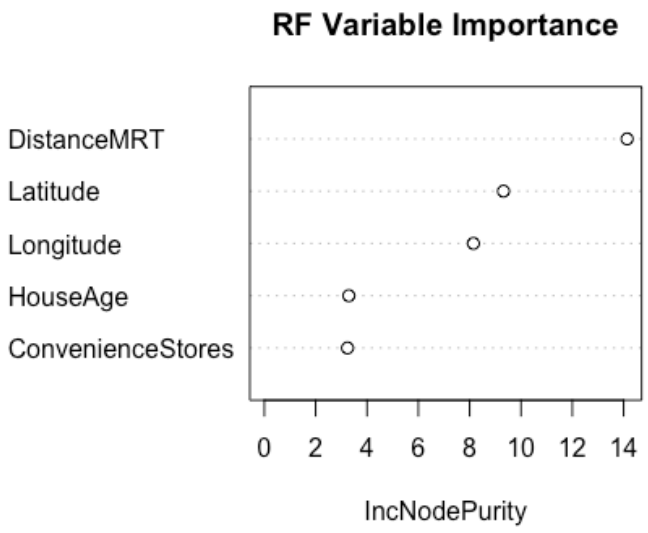

Figure 10.1. Random Forest variable importance.

10.3 Predicted vs Actual (Test)

The Predicted vs Actual scatterplot (Figure 10.2) compares predicted and observed log-prices on the test set. Most points lie close to the 45° reference line, indicating good predictive alignment. Performance is strongest in the central price range, while errors increase in the lower and upper tails, where fewer observations are available.

Overall, the Random Forest successfully captures the main pricing dynamics, although prediction uncertainty grows at the extremes.

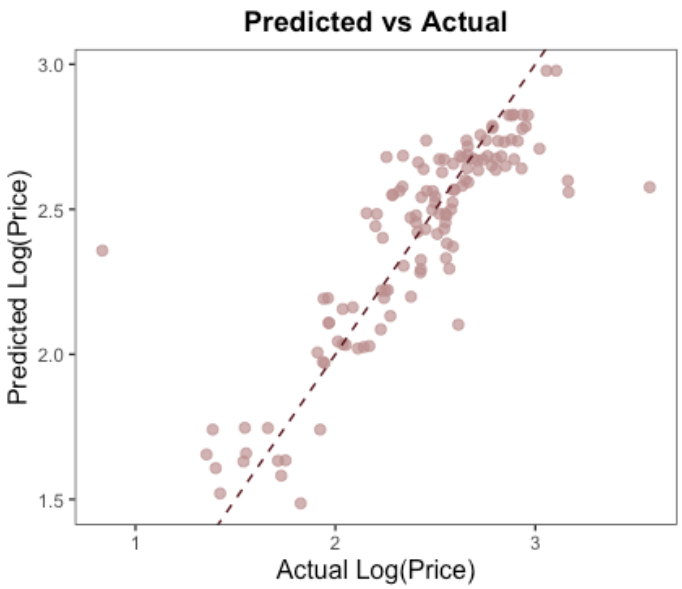

Figure 10.2. RF: Predicted vs Actual log-price (Test set).

11. Model Comparison

Performance is evaluated in terms of RMSE (lower = better) and R^2 (higher = better).

Table 11.1. Tuned Random Forest Performance

Model	R^2 Test	RMSE Test
Random Forest	**0.703**	**0.238**
Neural Network (MLP)	0.659	0.256
Linear Regression	0.604	0.275

1. **Random Forest is the best-performing model.**

   It yields the lowest RMSE (0.238) and the highest R^2 (0.703).

   This highlights that nonlinearities and interactions among predictors are important for predicting log-price in this dataset.

2. **Neural Network performs reasonably well.**

   Its RMSE (0.256) is close to the tuned RF.

   However, it does not surpass tree-based methods, likely because the dataset is small and relatively low-dimensional.

3. **Linear regression is less accurate.**

   This model fails to capture the complex spatial relationships present in the data.

12. Conclusion

This study examined the housing market of Taipei using unsupervised and supervised learning methods applied to the *Real Estate Valuation* dataset. The results reveal clear structure in the data, both in terms of market segmentation and price determinants.

Unsupervised methods (PCA and clustering) identify three distinct groups of properties, primarily differentiated by proximity to MRT stations, availability of convenience stores, and housing age. These patterns suggest that accessibility and neighborhood context play a central role in shaping market segmentation.

Supervised learning methods provide further insights into price formation. While linear regression captures general trends, it underperforms compared with more flexible models. Random Forest achieves the best predictive performance, outperforming both neural networks and parametric baseline. This performance gap highlights the presence of nonlinear spatial relationships and complex interactions that tree-based models are particularly suited to capture.

Random Forest also offers interpretable variable-importance measures, which confirm that distance to MRT stations and geographical coordinates are the strongest predictors of housing prices, while structural features such as house age are of more limited relevance. The neural network performs reasonably well, but it does not surpass Random Forest—likely due to the small–medium sample size and the ability of hierarchical splits to effectively capture dominant patterns without requiring deeper feature abstraction.

Overall, the evidence indicates that Taipei's housing market is characterized by

- nonlinear spatial determinants of price
- meaningful market segmentation.
- strong price variability across areas

These findings underscore the importance of incorporating flexible machine learning approaches when modeling urban housing prices, especially in environments where accessibility and location are key drivers of value.